Where might quantum regression models fit?

A benchmark across tides, wind, soccer and galaxy redshifts

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Abstract

Variational circuits that re-upload their inputs compute truncated Fourier series, which suggests they might be well suited to data whose inputs are genuinely periodic. We test this idea on four public regression problems that span the range from periodic to purely tabular inputs: tidal water levels, wind-turbine power, the probability that a soccer shot is scored, and galaxy redshift from photometry. A re-uploading circuit and a fidelity quantum kernel are compared with five classical learners and with the methods practitioners in each field use, under an equal hyperparameter budget, structure-aware data splits and training sets of 50 to 1000 points, for 850 runs in total. No quantum model outperforms the best classical or domain method on any dataset at any training size. The quantum kernel with an angle feature map is nonetheless competitive on the tidal and redshift problems, whereas the IQP feature map is consistently rejected on periodic inputs because its product terms break periodicity. A spectral analysis of the tidal problem shows that the frequencies available to a single-layer circuit already permit a test R^2 of 0.932, on par with the best classical model. The variational model's shortfall therefore stems from optimization and from the restricted set of Fourier coefficients the ansatz can reach, not from its frequency spectrum, while deeper circuits introduce frequencies the data do not contain and train less well.

1 Introduction

Whether quantum machine learning (QML) models offer any practical benefit on classical data remains an open question. Systematic benchmarks have found that, used out of the box, popular quantum classifiers rarely outperform simple classical baselines [1]. Rather than asking whether quantum models are better in general, it may be more productive to ask on which kinds of data their inductive bias is appropriate [2].

For the most widely used model class this bias is well characterized. A circuit that encodes each input feature through Pauli rotations, repeated L times, computes a Fourier series in the inputs whose frequencies are bounded by L [3, 4], and quantum kernels built from the same encodings are the corresponding kernel methods [5]. A natural hypothesis follows: quantum models should be most competitive when the inputs are angles and the target is a smooth periodic function of them.

In this work we test that hypothesis on regression problems drawn from four fields and chosen to span the axis from periodic to non-periodic inputs. Tidal water levels are, to a good approximation, a sum of cosines of astronomical phases; wind-turbine power depends on wind direction, an angle, as well as on wind speed; the probability that a soccer shot is scored depends on shot geometry, part of which is angular; and photometric redshift estimation has no periodic input at all and serves as a control. In each field we include the method a practitioner would actually use, so that the comparison reflects practice rather than a generic baseline.

Our contributions are as follows. (i) We provide a regression benchmark on four public real-world datasets in which every tunable model receives the same tuning budget, cross-validation respects temporal and grouped structure, and periodic inputs are encoded equivalently for quantum and classical models. (ii) We introduce a simple spectral analysis that separates what a circuit *can* express from what it *learns*, by comparing least-squares ceilings over a frequency set with the exact Fourier spectrum of the trained model. (iii) We report three findings: the angle-map quantum kernel is competitive on periodic and smooth tabular data; the IQP feature map is ill-suited to periodic inputs; and on the tidal problem the variational model is limited by trainability and coefficient reachability, not by its frequency spectrum.

2 Background

2.1 Re-uploading circuits as Fourier series

Consider d input angles $x \in [-\pi, \pi)^d$ encoded on d qubits by $S(x) = \bigotimes_{j=1}^{d} R_Y(x_j)$, interleaved with trainable unitaries $W_\ell(\theta_\ell)$:

$$U(x, \theta) = W_L S(x) \cdots W_1 S(x) W_0. \tag{1}$$

For an observable M, the model output is

$$f_\theta(x) = \langle 0 | U^\dagger(x, \theta)\, M\, U(x, \theta) | 0 \rangle = \sum_{\omega \in \Omega_L} c_\omega(\theta)\, e^{i\omega \cdot x}, \tag{2}$$

with integer frequencies $\Omega_L = \{-L, \ldots, L\}^d$ [3]. The encoding fixes the accessible frequencies, while the trainable blocks determine the coefficients c_ω. A given ansatz generally cannot realize every coefficient vector, so expressivity has two aspects: the spectrum Ω_L and the set of reachable coefficients.

2.2 Quantum kernels

A feature map $|\phi(x)\rangle = V(x)|0\rangle$ induces the fidelity kernel $k(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$ [5, 6]. For the angle map $V = S$ the kernel factorizes as

$$k(x, x') = \prod_{j=1}^{d} \cos^2\left(\frac{x_j - x'_j}{2} \right), \tag{3}$$

whose spectrum is exactly Ω_1. The IQP map [6] adds entangling ZZ rotations by products $x_j x_k$ and is a common default in QML software. The bandwidth of quantum kernels must be tuned for them to generalize [7].

2.3 Classical surrogates

Because Eq. (2) is a Fourier series, random Fourier features [8] can approximate such models classically [9]. We include them, together with a Gaussian process, as classical models with closely related inductive biases.

3 Benchmark design

3.1 Datasets

Table 1 summarizes the four datasets.

Table 1. Datasets. All are publicly available; N is the number of examples after cleaning. The test set is fixed for each domain and capped at 2000 examples.

Domain	Source	N	Inputs (type)	Target; split
Tides	NOAA CO-OPS [10], San Francisco, hourly, 2021–2023	26 280	phases of M2, S2, N2, K1, O1 (angle)	water level (m); test year 2023
Wind	Kelmarsh turbine 1 [11], 10-min SCADA, 2017	51 469	wind speed (real), direction (angle)	power / rated; test Oct–Dec
Soccer	StatsBomb open data [12], 198 matches	4 709	distance, goal angle (real); bearing (angle); 4 flags (binary)	goal (0/1); by match
Photo-z	SDSS DR18 [13], galaxies with $r < 19$, $z < 1$	50 693	$u-g$, $g-r$, $r-i$, $i-z$, r (real)	redshift; random

Tides. Classical harmonic analysis represents the water level as a sum of cosines of constituent phases $\phi_k(t) = \omega_k t \bmod 2\pi$ [14, 15]. We use the phases of the five largest constituents as inputs, so the inputs are exactly angles and the target is close to a low-order Fourier series in them. Meteorological surge cannot be predicted from time alone and sets an irreducible error floor. Models are trained on 2021–2022 and tested on 2023.

Wind. Turbine power is governed mainly by wind speed through the power curve, with smaller dependence on direction through wake and terrain effects. We keep intervals with full availability and no curtailment, normalize power by the rated 2050 kW, and test on the final quarter of 2017.

Soccer. Expected-goals (xG) models estimate the probability that a shot is scored. We use non-penalty shots from the 2022 World Cup, Euro 2020, Euro 2024 and Copa América 2024. The bearing of the shot is an angle; distance and the angle subtended by the goal are not. Splits are by match, so that no match contributes to both training and test sets.

Photometric redshift. Estimating galaxy redshift from broadband colors is a standard tabular regression problem with no periodic input, and serves as our control.

3.2 Input encoding

Each feature is typed as an angle, a real number or a binary flag, and both model families receive equivalent information about periodicity. Quantum models take angle features unchanged in $[-\pi, \pi)$, so that wrap-around is exact. Real features are mapped to the half circle $[-\pi/2, \pi/2]$ using the 0.5% and 99.5% quantiles of the training data, with clipping; mapping to the full circle would identify the two extremes of a non-periodic variable. Binary features are mapped to $\pm\pi/2$. Classical models receive $(\sin x_j, \cos x_j)$ for each angle feature and standardized real features.

3.3 Models

Re-uploading circuit. We use one qubit per feature, R_Y encoding, L blocks of strongly entangling layers and an initial trainable block W_0. The prediction is $a\bar{z} + b$, where $\bar{z}$ is the mean of the single-qubit $\langle Z_i \rangle$ and a, b are trainable; for the soccer task a sigmoid is applied. Training uses Adam on the mean squared error (binary cross-entropy for soccer) with standardized targets, mini-batches of 128, at most 400 epochs, and early stopping with patience 40 on a 15% validation split. The depth $L \in \{1, \ldots, 4\}$ and the learning rate are tuned.

Quantum kernel. The feature map (angle or IQP), the number of IQP repetitions, a band-

width applied to non-angle inputs and the regularization strength are tuned. We use kernel ridge regression for regression tasks and ℓ_2-regularized logistic regression on kernel features for soccer. In simulation each state is prepared once per input, so the Gram matrix costs $O(n)$ circuit evaluations; on hardware, overlap estimation would require $O(n^2)$.

Classical models. Ridge regression (logistic regression for soccer), random Fourier features with ridge regression, a multilayer perceptron, histogram gradient boosting and a Gaussian process with an RBF and white-noise kernel whose hyperparameters are set by maximizing the marginal likelihood, all implemented in scikit-learn [16].

Domain methods. For tides we use UTide harmonic analysis with eight constituents and nodal corrections [15]; for wind, a binned power curve in the spirit of IEC 61400-12-1 [17]; for soccer, logistic regression on shot distance and goal angle, the textbook xG model; and for redshift, distance-weighted k-nearest neighbors [18]. These methods are fitted on the same training points as every other model. We also report two *published references*, NOAA's official tide prediction and StatsBomb's xG model. Both were fitted on far more data, so we treat them as practical ceilings rather than competitors.

3.4 Protocol

For every dataset, model, training-set size $n \in \{50, 100, 250, 500, 1000\}$ and seed $s \in \{0, \dots, 4\}$, we sample n training points from a fixed training pool and evaluate on a fixed test set. Each tunable model receives ten trials of Optuna's TPE sampler [19]. Each trial is scored by threefold cross-validation on the training points alone, with contiguous time blocks for tides and wind and folds grouped by match for soccer. The selected configuration is refitted on all n points. The search spaces are listed in Appendix A. We report R^2 for the regression tasks and the Brier skill score (BSS), $1 - \text{Brier}/\text{Brier}_{\text{base rate}}$, for soccer. Circuits are simulated without noise in PennyLane [20].

3.5 Spectral analysis

For the purely angular tides data, two quantities can be computed exactly. The *spectrum ceiling* of a frequency set Ω is the test score of the least-squares fit of the target onto $\{1, \cos(\omega \cdot x), \sin(\omega \cdot x) : \omega \in \Omega\}$, using the full training pool of 17 520 hours. No model whose spectrum is confined to Ω can be expected to do better. The *model spectrum* is obtained by evaluating a trained model on a regular grid of $(2K + 1)^d$ points and applying a d-dimensional fast Fourier transform. For a band-limited model such as the re-uploading circuit this recovers the coefficients c_ω of Eq. (2) exactly. The Fourier coefficients of the *data*, in contrast, are not identifiable: all samples lie on a single line in time, and tidal frequencies are integer combinations of a few astronomical ones (for example, $\omega_{\text{K1}} + \omega_{\text{O1}} = \omega_{\text{M2}}$). We therefore compare only nested frequency sets.

4 Results

4.1 Overall comparison

Table 2 and Fig. 1 summarize the main result: no quantum model is the best-performing model on any dataset or at any training-set size (see also Table 6). The quantum kernel comes closest. With 1000 training points it reaches $R^2 = 0.927$ on tides, against 0.931 for the best classical model and 0.930 for harmonic analysis, and exceeds NOAA's official prediction (0.925) for the test year. On the redshift control it reaches 0.902, against 0.909. On wind, where the target is dominated by the power curve, all flexible models perform similarly and the domain method reaches 0.980. The

Table 2. Test scores with $n = 1000$ training points: mean ± standard deviation over five seeds (higher is better). Bold marks the best model trained on the same points. R^2 for regression; BSS is the Brier skill score relative to the base rate, for the probability task.

	Model	Tides (R^2)	Wind (R^2)	Soccer (BSS)	Photo-z (R^2)
Classical	Ridge / logistic	0.899 ± 0.002	0.900 ± 0.004	$\mathbf{0.091 \pm 0.009}$	0.818 ± 0.010
	Random Fourier features	0.926 ± 0.007	0.979 ± 0.001	0.084 ± 0.008	$\mathbf{0.909 \pm 0.006}$
	Multilayer perceptron	0.926 ± 0.003	0.977 ± 0.004	-0.076 ± 0.239	0.863 ± 0.080
	Gradient boosting	0.912 ± 0.003	0.981 ± 0.001	0.075 ± 0.010	0.904 ± 0.005
	Gaussian process	$\mathbf{0.931 \pm 0.002}$	$\mathbf{0.981 \pm 0.001}$	0.090 ± 0.006	0.908 ± 0.022
Quantum	Re-uploading circuit	0.894 ± 0.013	0.976 ± 0.006	0.061 ± 0.023	0.893 ± 0.007
	Quantum kernel	0.927 ± 0.003	0.953 ± 0.018	0.087 ± 0.008	0.902 ± 0.005
Domain	Domain method[a]	0.930 ± 0.002	0.980 ± 0.001	0.066 ± 0.003	0.882 ± 0.011
	Published reference[b]	0.925	—	0.149	—

[a] UTide harmonic analysis (tides), binned power curve (wind), distance–angle logistic regression (soccer) and k-nearest neighbours (photo-z), fitted on the same training points as every other model. [b] NOAA's official tide prediction and StatsBomb's expected-goals model; both are fitted on far more data and mark a practical ceiling rather than a competitor.

Table 3. Spectrum ceilings for tides: test R^2 of the least-squares fit on each frequency set, using all training hours. $\Omega_1^{\text{add}} \subset \Omega_1$ contains only frequencies with a single nonzero component.

Set	Model with this spectrum	Frequencies	Ceiling R^2
Ω_1^{add}	single-feature terms only	5	0.899
Ω_1	one-layer circuit, angle kernel	121	0.932
Ω_2	two-layer circuit	1,562	0.941

re-uploading circuit reaches 0.976 there but falls behind on tides (0.894). On soccer the best trained models attain a Brier skill score of about 0.09, well below StatsBomb's xG (0.149), which draws on information absent from our features, such as the positions of defenders and the goalkeeper. At small training-set sizes the Gaussian process and the domain methods are the most reliable; the quantum kernel needs a few hundred points to approach them, and the variational circuit and the multilayer perceptron vary most across seeds.

4.2 Choice of feature map

The tuner was free to choose between the angle and IQP feature maps. It selected the angle map in 25 of 25 tides runs and 25 of 25 wind runs, but in only 10 of 25 redshift runs. The explanation is structural. The IQP map rotates qubit pairs by products $x_j x_k$; for angles spanning $[-\pi, \pi)$ these products are not 2π-periodic in either argument, so the resulting kernel does not respect the periodicity of the inputs. Without tuning, the IQP kernel achieved $R^2 \leq 0$ on tides. Rescaling the inputs cannot repair this without destroying the periodicity of each angle, so the commonly used IQP map is a poor default for periodic data.

4.3 Spectral analysis of the tidal problem

The tuned circuit used a single layer in 25 of 25 tides runs and remained below ridge regression, which raises the question of whether a single layer simply lacks the frequencies the tides require. The ceilings in Table 3 rule this out. The one-layer spectrum Ω_1 supports a test R^2 of 0.932, equal to the best model in the benchmark, and enlarging it to Ω_2 adds less than 0.01. What matters

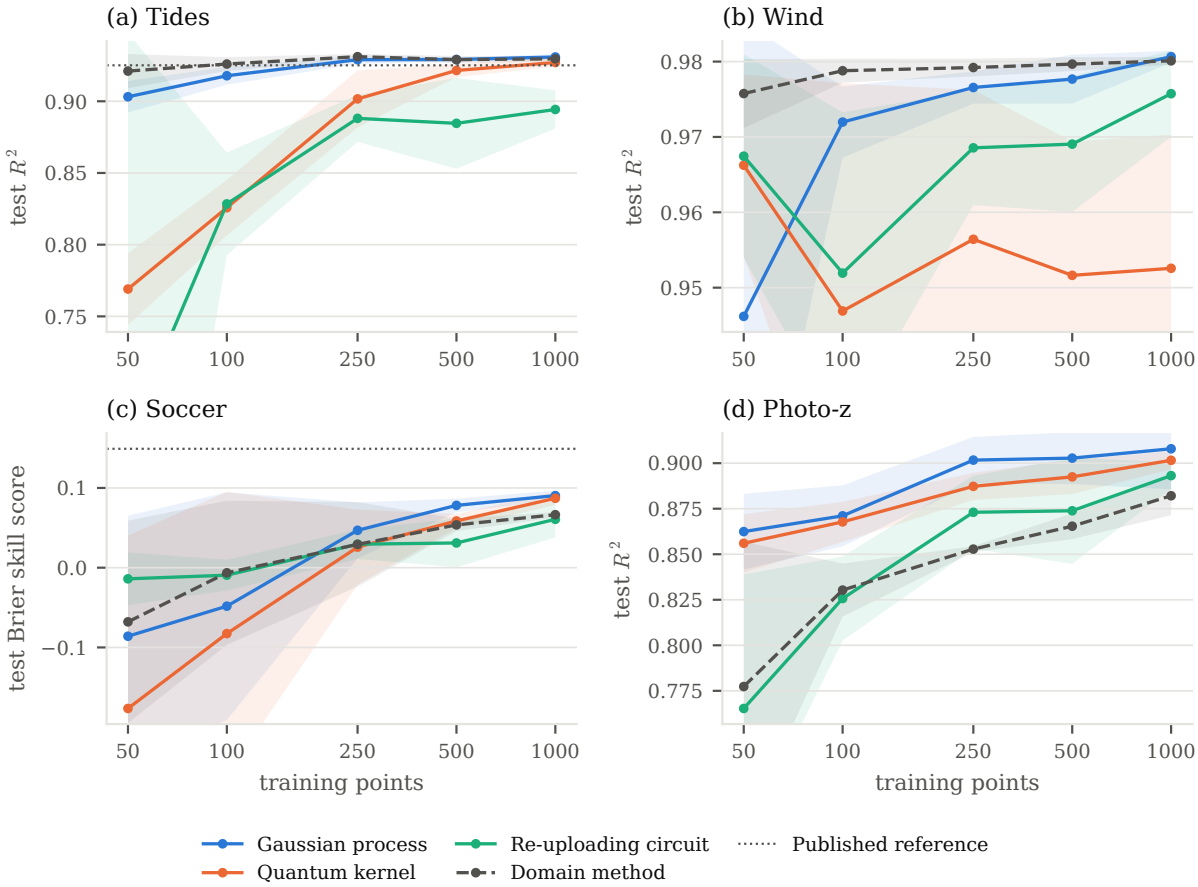


Figure 1. Test score versus training-set size for the best classical model (Gaussian process), the two quantum models and the domain method. Lines show the mean and shaded bands one standard deviation over five seeds; dotted lines mark published references. Points below the plotted range are clipped.

is not higher harmonics but interactions: restricting Ω_1 to single-feature frequencies lowers the ceiling to 0.899, because combinations of the five input phases reproduce constituents that are not themselves inputs. The angle kernel of Eq. (3), whose spectrum is exactly Ω_1, is therefore well matched to the problem, consistent with its performance (Fig. 2).

Table 4 locates the shortfall of the variational model. With a short training budget, early stopping ends training after 23 epochs and the circuit reaches a test R^2 of 0.833. With a longer budget, the same circuit reaches 0.905. Even when trained on a noise-free target that lies inside its spectrum, the circuit recovers only about 97% of the target's variance: Ω_1 comprises 243 real Fourier coefficients, whereas the one-layer ansatz has 32 parameters and cannot set them independently. Deeper circuits perform worse rather than better. With $L = 2$ or 3, roughly 30% of the model's variance lies at frequencies with $\max_j |\omega_j| = 2$, which the data do not require, and training R^2 falls to about 0.7. In every case the model spectrum vanishes above L to numerical precision, as Eq. (2) predicts.

The full benchmark confirms the role of training. A first run of the benchmark used the short budget; repeating every variational run with the longer budget described in Section 3.3 raised the circuit's mean tides score at $n = 1000$ from 0.870 to 0.894 and halved its spread across seeds (from 0.026 to 0.013), with similar gains at small training-set sizes on wind and redshift and little change

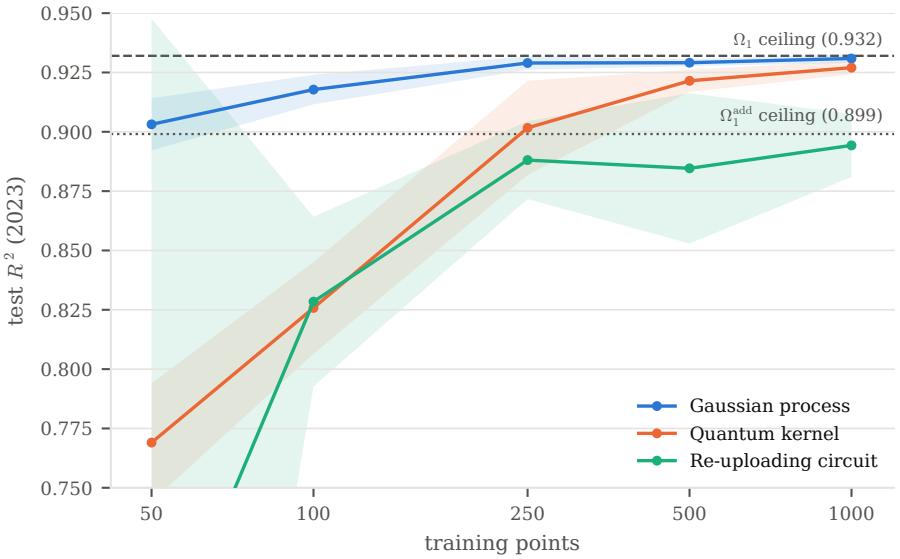


Figure 2. Tides: models compared with the spectrum ceilings of Table 3. The one-layer ceiling is reached by the classical and kernel models but not by the variational circuit.

Table 4. Re-uploading circuits trained on tides ($n = 1000$, seed 0). The short budget is 100 epochs with patience 15; the long budget is 600 epochs with patience 60. The *teacher* target is the noise-free least-squares fit on Ω_1, against which its R^2 is measured. P_2 is the share of the model's variance at frequencies with $\max_j |\omega_j| = 2$.

L	Budget	Target	Epochs	Train R^2	Test R^2	P_2 (%)
1	short	data	23	0.891	0.833	0
1	long	data	161	0.941	0.905	0
2	long	data	86	0.701	0.563	32
3	long	data	117	0.675	0.544	29
1	long	teacher	375	0.971	0.967	0
2	long	teacher	164	0.705	0.592	34

on soccer. The circuit nevertheless remains below both the ceiling and ridge regression. Its final fits on tides ran for a median of 373 of the 400 permitted epochs, so it was most likely still improving when training stopped.

4.4 Computational cost

Training the re-uploading circuits accounted for 19.5 of the 19.8 CPU-hours used by the entire benchmark. The quantum kernels required only minutes, because states were simulated once per input.

5 Discussion

Our results support the periodic-input hypothesis only in a weak form. The angle-map quantum kernel is a sound model for angular data: its spectrum matches what the tidal problem requires, and it is competitive with the best classical models. However, that kernel is the simple, classically computable product of Eq. (3), so its success is not evidence of a quantum advantage. The variational circuit has an equally appropriate spectrum but is harder to train, cannot reach every

coefficient within that spectrum, and is harmed by the additional frequencies that deeper circuits introduce. The spectral analysis of Section 3.5 makes these distinctions measurable: comparing a ceiling with a model's score separates limits of the spectrum from limits of training or of the ansatz.

Several limitations qualify these conclusions. All circuits were simulated without hardware or shot noise. Simulation cost restricted training sets to at most 1000 points and tuning to ten trials per run. The tidal analysis uses a single station and a single test year. The soccer data combine four tournaments, and the spectroscopic redshift sample is biased toward bright galaxies, as is common for photometric-redshift training sets. The variational circuits were limited to 400 training epochs, which most tides fits nearly exhausted. Finally, the probes in Table 4 use a single seed, so their exact values are indicative only.

6 Conclusion

On four real regression problems, under an equal tuning budget, we find no advantage for quantum models. We do find that the Fourier picture of quantum models is predictive in practice: it explains which feature map suits which data, and it separates expressivity from trainability as the reason a variational circuit falls short. Natural next steps are to assess robustness to shot noise, to extend the tidal analysis to other stations, and to use spectrum ceilings to guide the design of circuits and feature maps before training.

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Data and code availability

All data are public (Table 1). The code, configuration files and scripts that produce every table, figure and number in this paper are available at *[repository link]*.

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A Hyperparameter search spaces

Table 5. Hyperparameters tuned for each model, with ten trials per run. UTide, the logistic xG model and the published references have no tuned hyperparameters.

Model	Hyperparameter	Range
Ridge / logistic	`alpha`	$[10^{-4}, 100]$, log
Random Fourier features	`n_components`	$[50, 500]$, integer
	`gamma`	$[0.01, 10]$, log
	`alpha`	$[10^{-4}, 100]$, log
Multilayer perceptron	`hidden`	$[8, 128]$, integer
	`layers`	$[1, 3]$, integer
	`alpha`	$[10^{-6}, 0.1]$, log
	`lr`	$[10^{-4}, 0.01]$, log
Gradient boosting	`max_depth`	$[2, 8]$, integer
	`lr`	$[0.01, 0.3]$, log
	`max_iter`	$[50, 500]$, integer
Gaussian process	none (marginal likelihood)	
Re-uploading circuit	`n_layers`	$[1, 4]$, integer
	`lr`	$[10^{-3}, 0.1]$, log
Quantum kernel	`feature_map`	{angle, iqp}
	`reps`	$[1, 3]$, integer
	`bandwidth`	$[0.05, 2]$, log
	`alpha`	$[10^{-6}, 1]$, log
Wind: power curve	`bin_width`	$[0.25, 1.5]$, linear
Photo-z: knn	`k`	$[3, 50]$, integer

B Results with 100 training points

Table 6. Test scores with $n = 100$ training points: mean $\pm$ standard deviation over five seeds (higher is better). Bold marks the best model trained on the same points. R^2 for regression; BSS is the Brier skill score relative to the base rate, for the probability task.

	Model	Tides (R^2)	Wind (R^2)	Soccer (BSS)	Photo-z (R^2)
Classical	Ridge / logistic	0.891 ± 0.012	0.889 ± 0.018	$\mathbf{0.010 \pm 0.076}$	0.772 ± 0.055
	Random Fourier features	0.894 ± 0.024	0.959 ± 0.027	-0.021 ± 0.114	0.803 ± 0.098
	Multilayer perceptron	0.880 ± 0.011	0.928 ± 0.023	-0.564 ± 0.505	-0.268 ± 1.376
	Gradient boosting	0.838 ± 0.020	0.825 ± 0.050	-0.015 ± 0.045	0.835 ± 0.008
	Gaussian process	0.918 ± 0.006	0.972 ± 0.005	-0.048 ± 0.143	$\mathbf{0.871 \pm 0.017}$
Quantum	Re-uploading circuit	0.828 ± 0.036	0.952 ± 0.021	-0.009 ± 0.019	0.826 ± 0.023
	Quantum kernel	0.826 ± 0.019	0.947 ± 0.030	-0.083 ± 0.178	0.868 ± 0.011
Domain	Domain method[a]	$\mathbf{0.926 \pm 0.005}$	$\mathbf{0.979 \pm 0.002}$	-0.006 ± 0.090	0.830 ± 0.015
	Published reference[b]	0.925	—	0.149	—

[a] UTide harmonic analysis (tides), binned power curve (wind), distance–angle logistic regression (soccer) and k-nearest neighbours (photo-z), fitted on the same training points as every other model. [b] NOAA's official tide prediction and StatsBomb's expected-goals model; both are fitted on far more data and mark a practical ceiling rather than a competitor.